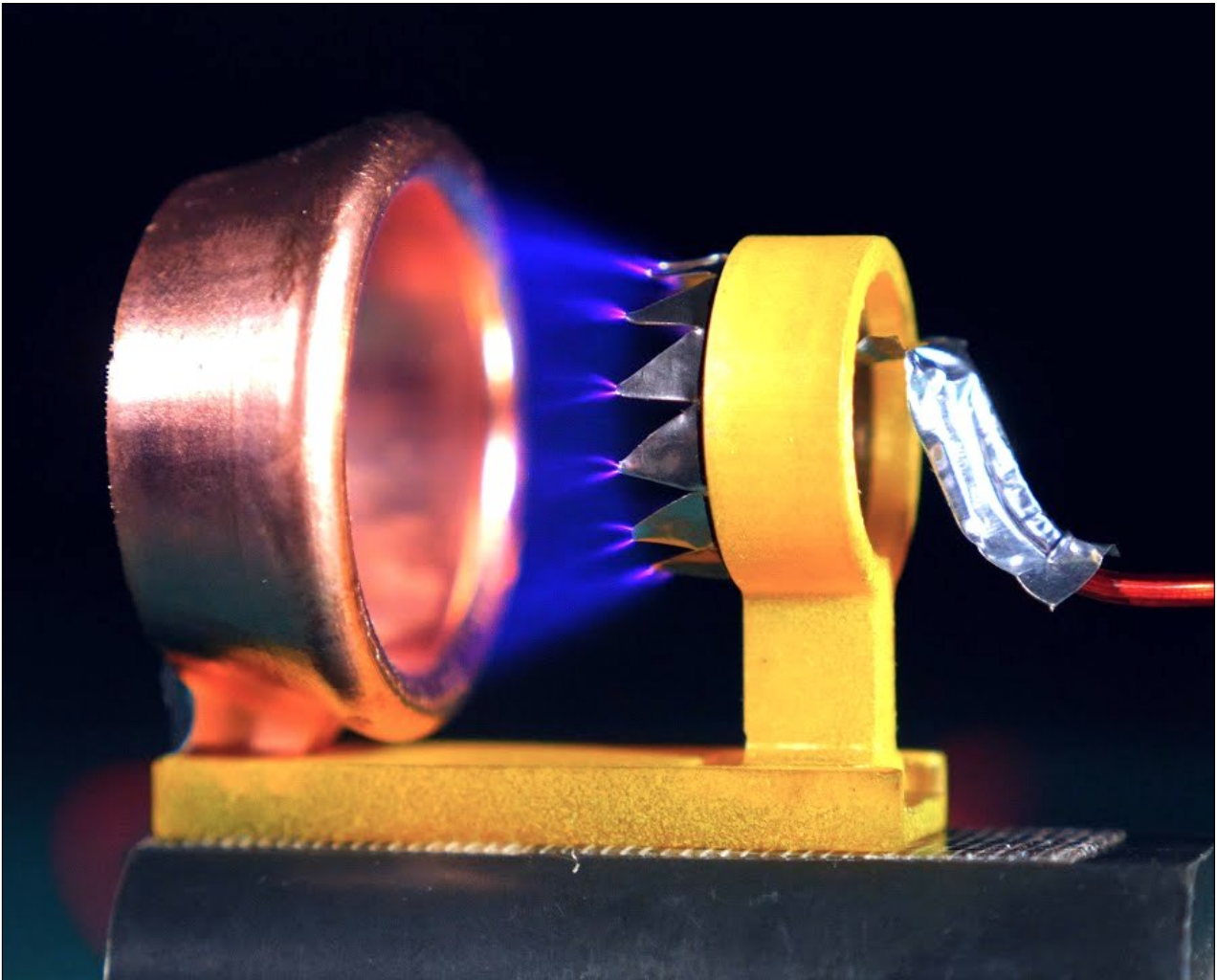




Ion Thruster Model

Prepared for: DBL Science Exhibition (Physics)

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3 November 2024

SYNOPSIS

Objective

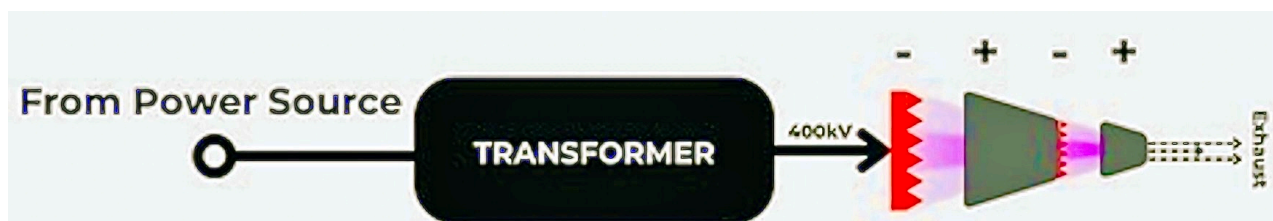
Create a model electric propulsion system for spacecraft

Principle

Ion thrusters create thrust by expelling beams of electrically charged ions, using the conservation of momentum principle to propel a spacecraft forward. By taking advantage of the charge-to-mass ratio of ions, these thrusters can reach high exhaust velocities even with relatively low potential differences. This makes them highly efficient. The 400kV transformer used in an ion thruster model - amplifies this effect by generating a strong electric field, which ionises and accelerates particles, typically from a noble gas like xenon. As ions are stripped of electrons, they become positively charged and respond to the electric field, reaching high speeds as they exit the thruster, thereby producing thrust.

This approach has the advantage of reducing the reaction mass, or fuel, needed compared to chemical rockets. This allows for higher specific impulses, meaning the thrust is maintained over a longer period. However, this efficiency comes with a trade-off. The low thrust-to-power ratio means that ion thrusters are not suited for launching spacecraft from Earth's surface, as they generate lower acceleration than chemical rockets. Instead, ion thrusters are ideal for long-duration in-space missions, where their continuous, efficient propulsion can steadily accelerate spacecraft over vast distances, making them valuable for deep-space exploration.

Process Description



Ion Thruster Model (Flow Diagram)

In an ion thruster powered by a 400kV transformer, thrust is generated through a simple but powerful setup that relies solely on electric current—no separate fuel supply is needed. The high-voltage transformer creates an intense electric field. As the transformer powers up, this electric field interacts with the surrounding air, stripping electrons from oxygen and nitrogen molecules and creating positively charged ions right from the

ambient air. This process, known as ionisation, is the key to making the thruster work without additional propellant.

Once the air molecules become ions, they respond to the electric field, accelerating away from the positively charged electrode. This movement creates a directional flow of ions that propels the thruster forward. As the ions are expelled, they generate a gentle but continuous thrust that can be adjusted by fine-tuning the voltage level of the transformer.

This design takes advantage of the natural environment, using the transformer's electric field to produce ions and thrust without the weight or complexity of a separate gas supply. While the thrust is low, this approach is highly efficient and ideal for certain applications where continuous, reliable propulsion is beneficial.

Resources

The ion thruster setup using a 400kV transformer relies on relatively accessible components and a simple design, making it more straightforward than chemical or traditional electric propulsion systems. By using ambient air as the "fuel," it bypasses the need for heavy, expensive propellant tanks and complex fuel delivery systems. The transformer generates the ionising electric field, reducing the complexity and weight of the setup. This thruster can continuously provide low-level thrust over extended periods, which is ideal for gradual acceleration, making it particularly resource-efficient in environments where the need for raw materials or complex engineering is minimised.

Limitations

Despite its simplicity and efficiency, the ion thruster powered by a transformer has notable limitations. The low thrust output means it's unsuitable for high-acceleration tasks, such as launching from Earth's surface, where powerful engines are required to overcome gravity quickly. This setup is also dependent on a high-voltage power source, which may not be ideal in remote applications without a consistent energy supply. Additionally, since it relies on ambient air for ionisation, it cannot function in a vacuum, limiting its usefulness in outer space. Thus, while it's resource-friendly and effective for specific in-atmosphere applications, it falls short for high-speed or space-bound missions.

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